BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 231 · Scope, modularity and parameters

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. If procedural Deposit(balance,5) returns a value that the caller ignores, does balance change?

Q2. What is a local variable?

Q3. How does a value-returning function pass a result back?

Q4. Why might using ref unexpectedly make a method harder to debug?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

scope — Where a name can be accessed.

local — Variable with block or method-level scope.

global/class-level — Variable accessible in a wider scope, subject to access rules.

by value — Pass a copy of the value.

by reference — Pass an alias of the caller's variable.

New learning

Copying an integer — Assignment b=a copies the integer 5, not a reference to a storage location.

Passing by reference — Increase(ref a) gives parameter value an alias of a, so the caller a changes to 8.

Use modular methods — A short reusable method has a clear input effect, can be tested, and reduces repeated code.

Worked example

Trace AddFour(x) by value when caller x starts at 6.

1. Predict first — The function parameter number receives a copy of 6.

2. Model and check — Inside the function number becomes 10; caller x remains 6.

3. Model and check — The parameter leaves scope at the end of the call.

We Do · Practise with guidance

Trace AddFour(ref y) when caller y starts at 9.

```
static void AddFour(ref int number) {
 number += 4;
}
int y=9; AddFour(ref y);
```

Predict first

The ref parameter number refers to y, not a fresh copy.

Model and check

Adding four changes caller y from 9 to 13.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Method AddTwo(int n) adds 2 to n. Caller x=7 is passed by value. What is x after the call?

Question 2

Write an OCR-style subroutine Swap with two by-reference integer parameters and explain why by-reference is required.

Question 3

Compare `int result=Double(x);` with `Double(ref x);` as ways of updating a number. Give advantages of one method for a pure calculation.

Exit Ticket · Answer before checking

Passing an integer by value in C# changes the caller's integer when the parameter changes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. No. The returned value is discarded.

Q2. A variable declared inside a method or block, with restricted scope.

Q3. Using return.

Q4. The method can change a caller's variable as a side effect, even if there is no returned result.

You Do pages · Independent questions

Q1. [3] n is a copy of 7 [1], becomes 9 inside method [1], caller x remains 7 [1].

Q2. [6] procedure Swap(byRef a,byRef b) [1]; temp=a [1]; a=b [1]; b=temp [1]; caller variables must be updated rather than local copies [2].

Q3. [6] Return version computes a result [1] and caller controls assignment [1]; ref version can update x in place [1]; returning avoids hidden state changes [1], is easier to reason about [1] and can be tested using input/output assertions [1].

Exit Ticket · Model answer

False. A copy is passed unless a reference-passing mechanism such as ref is used.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding